

Fmk6 Lec 11 Series 5

Review

$$\begin{aligned}
 1 \quad a \\
 2 \quad \log y &= 2 \log x + 0.6 \\
 &= \log x^2 + \log 10^{0.6} \\
 &= \log(10^{0.6} x^2) \\
 y &= 0.6 x^2
 \end{aligned}$$

$$\begin{aligned}
 3 \quad dr^2 &= 36 \\
 dr^4 &= 81 \\
 r^2 &= \frac{d}{4} \\
 r &= \frac{3}{2}
 \end{aligned}$$

$$a = 16$$

$$4 \quad a$$

$$5 \quad \frac{3x+1}{2x-1} = y$$

$$3x+1 = y(2-x)$$

$$3x+4x = 2y-1$$

$$x = \frac{2y-1}{y+3}$$

$$y = \frac{2x-1}{x+3} \quad (d)$$

Exam Qs

$$1 \quad a) (1+2x)^{\frac{1}{2}}$$

$$= 1 + x + \frac{\frac{1}{2} \cdot -\frac{1}{2}}{2!} (2x)^2 + \frac{\frac{1}{2} \cdot -\frac{1}{2} \cdot -\frac{3}{2}}{3!} (2x)^3$$

$$= 1 + x - \frac{1}{2}x^2 + x^3$$

$$\begin{aligned}
 6) (1+9x^2)^{-1} &= 1 - 9x^2 + \frac{-1 \cdot -2}{2!} (9x^2)^2 + \frac{-1 \cdot -2 \cdot -3}{3!} (9x^2)^3 \\
 &= 1 - 9x^2 + 81x^4 - 729x^6
 \end{aligned}$$

$$= (1 + x - \frac{1}{2}x^2 + x^3)(1 - 9x^2)$$

$$= 1 + x - \frac{1}{2}x^2 - 9x^2 + x^3 - 9x^3$$

$$= 1 + x - \frac{19}{2}x^2 - 8x^3$$

$$c) |x| < \frac{1}{3}, \quad x \neq -\frac{1}{9}$$

$$2) a) \frac{A}{(1-2x)} + \frac{B}{(2+x)} + \frac{C}{(2+x)^2} = \frac{5+4x-3x^2}{(1-2x)(2+x)^2}$$

$$A(2+x)^2 + B(1-2x)(2+x) + C(1-2x) = 5+4x-3x^2$$

$$5C = -25$$

$$C = -5$$

$$\frac{25}{4}A = \frac{25}{4}$$

$$A = 1$$

$$Aa^2 - 2Bx^2 = -3x^2$$

$$\therefore B = 2$$

$$\frac{1}{1-2x} + \frac{2}{2+x} - \frac{5}{(2+x)^2}$$

$$b) (1-2x)^{-1} + (1+\frac{1}{2}x)^{-1} - \frac{5}{4} \left(1+\frac{1}{2}x\right)^{-2}$$

$$= 1 + 2x + \frac{-1 \cdot 2}{2!} (2x)^2 + 1 - \frac{1}{2}x + \frac{-1 \cdot 2}{2!} \left(\frac{1}{2}x\right)^2 - \frac{5}{4} \left(1 - x + \frac{-2 \cdot 3}{2!} \left(\frac{1}{2}x\right)^2\right)$$

$$= 1 + 2x + 4x^2 + 1 - \frac{1}{2}x + \frac{1}{4}x^2 - \frac{5}{4} + \frac{5}{4}x - \frac{15}{16}x^2$$

$$= \frac{3}{4} + \frac{11}{4}x + \frac{93}{16}x^2$$

$$c) |x| < \frac{1}{2}$$